Smart Energy Management

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

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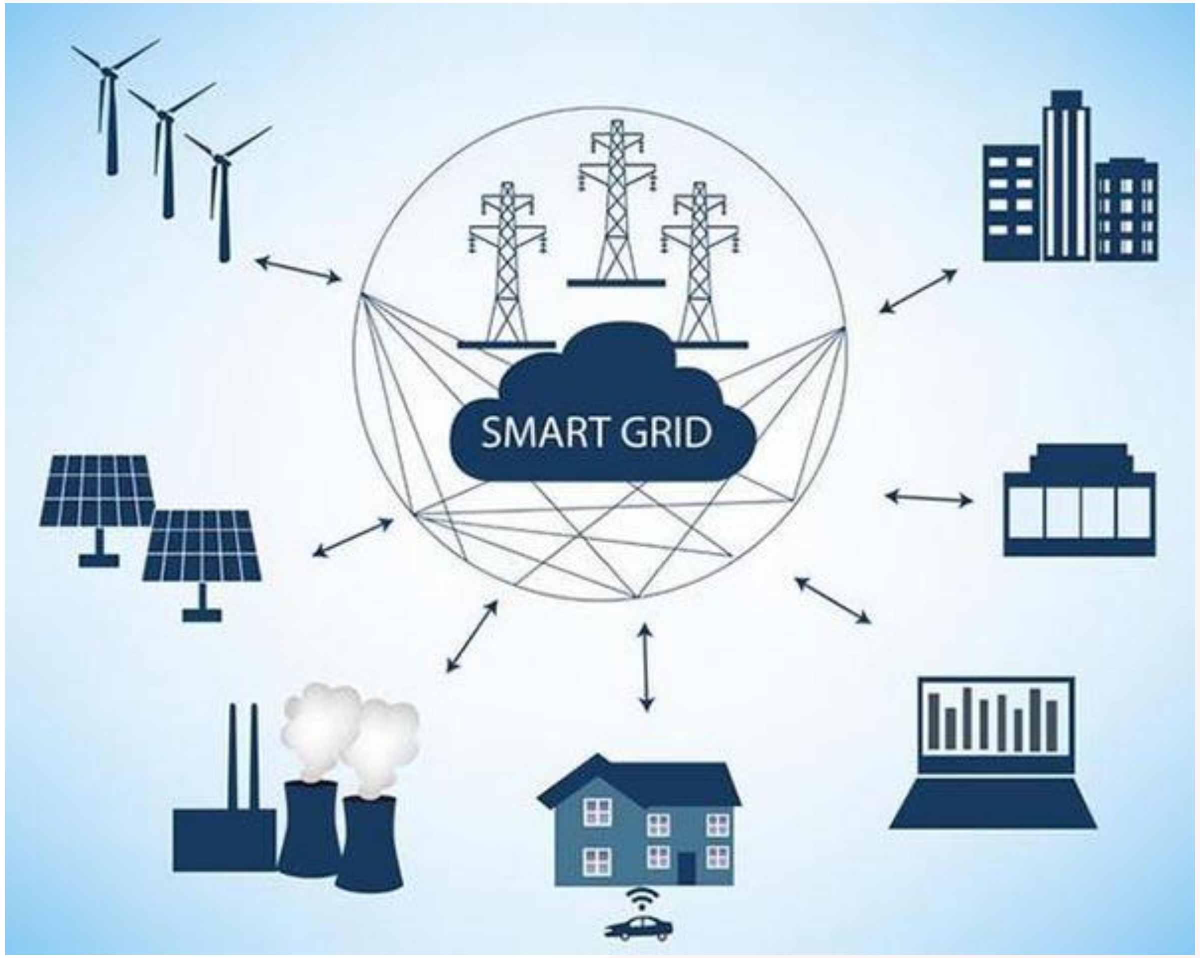
Introduction to Smart Grids

Smart Grids

- Electricity network that can intelligently integrate the actions of all users in a smart city
  - i.e., generators and consumers

- Smart grids efficiently deliver sustainable, economic, and secure electricity supplies



https://kvcable.com/it/global-smart-grid-market-to-grow-at-a-cagr-of-19-1-from-2021-26/

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Smart Grids (cont.)

- Smart grids leverages information technologies and communication for real-time applications
  - two-way exchange of information!

- Self-healing, adaptive, resilient and sustainable

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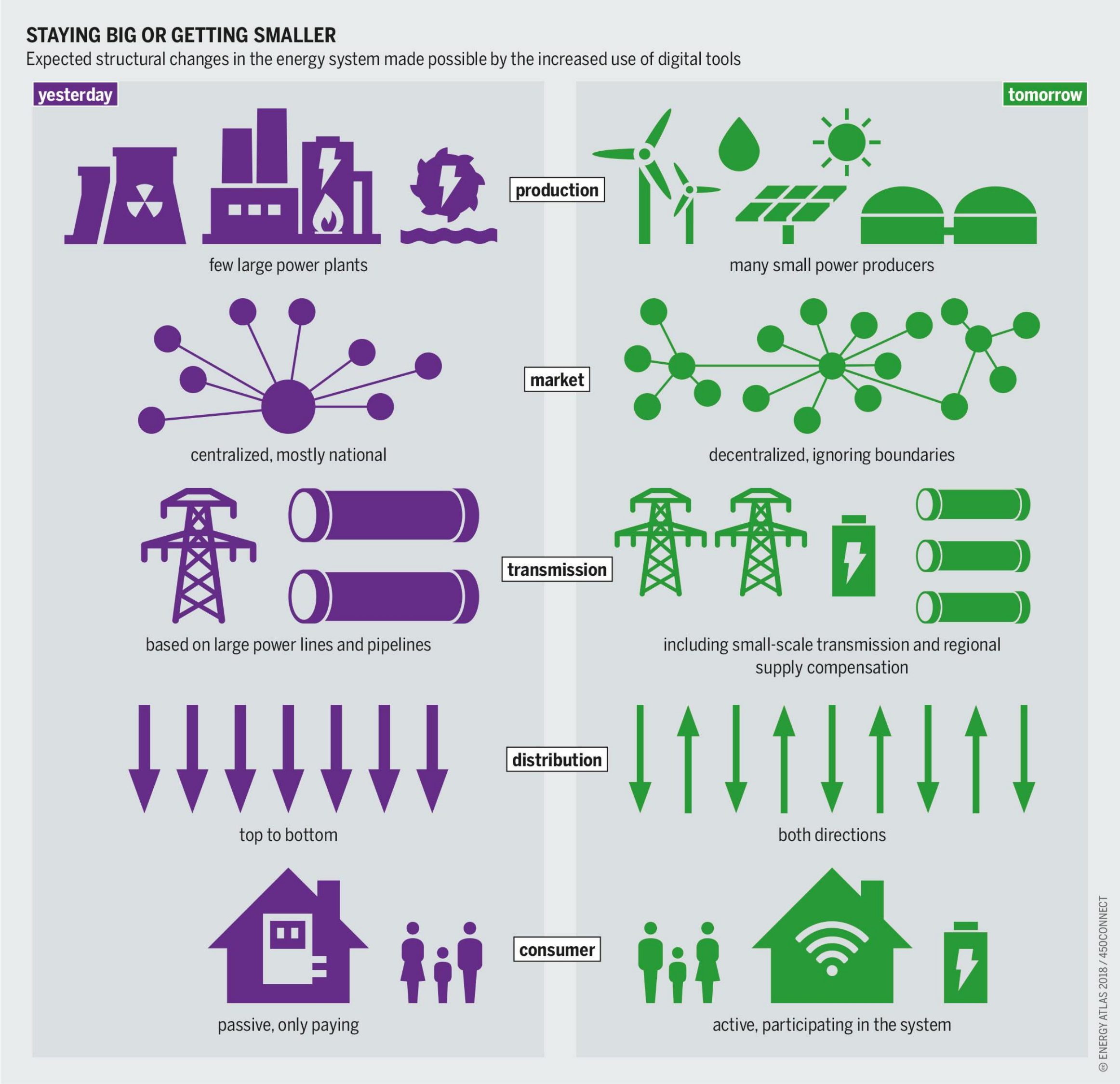
Traditional VS Smart Grid



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Smart Grid is the future!

- Current infrastructure are centralized and not optimized for specific needs
- The U.S. Department of Energy aims to have a fully functioning smart grid in place by 2035 to help cut carbon emissions.
- Smooth transition to smart energy management still requires research on algorithms and technology

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AI tasks in Smart Grids

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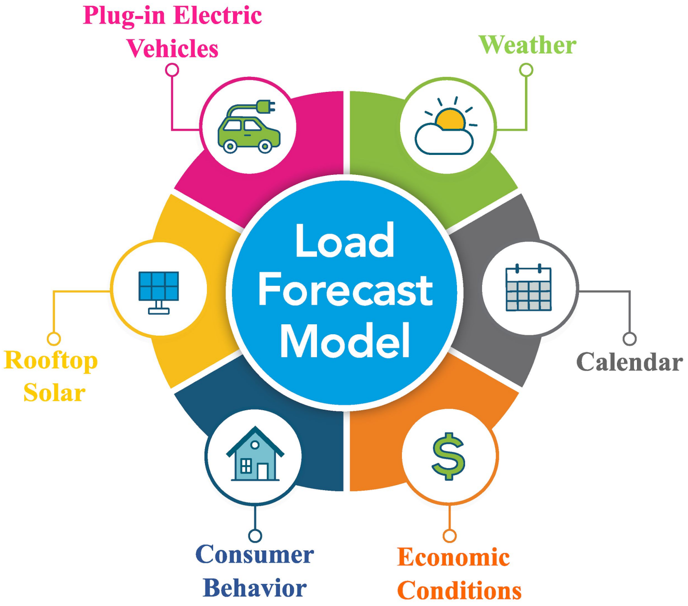
Using AI in Smart Grids

- AI is a powerful technology to transform traditional grids into "smart grids"
- Several important tasks in this area can indeed be tackled with advanced AI approaches
- Two types of approaches:
  - **Virtual AI:** analyzing data in smart grids to help grid operators or consumers (recommender systems)
  - **Physical AI:** self-aware AI that can optimize and control grid operations without human intervention

Load Forecasting

- Predicting energy consumption is important for many reasons:
  - to balance the electricity supply and demand
  - predicting peaks
  - plan future works on energy infrastructure
- Crucial to improve reliability, stability, reducing costs, and optimizing the infrastructure
- At different granularities:
  - short-term (e.g., from minutes to hours)
    - real-time control, energy transfer scheduling, demand response
  - mid-term (from hours to weeks)
    - optimizing load dispatch, maintenance scheduling, balancing demand and generation
  - long-term (years)
    - to predict the power consumption, system planning, and scheduling of system's expansions



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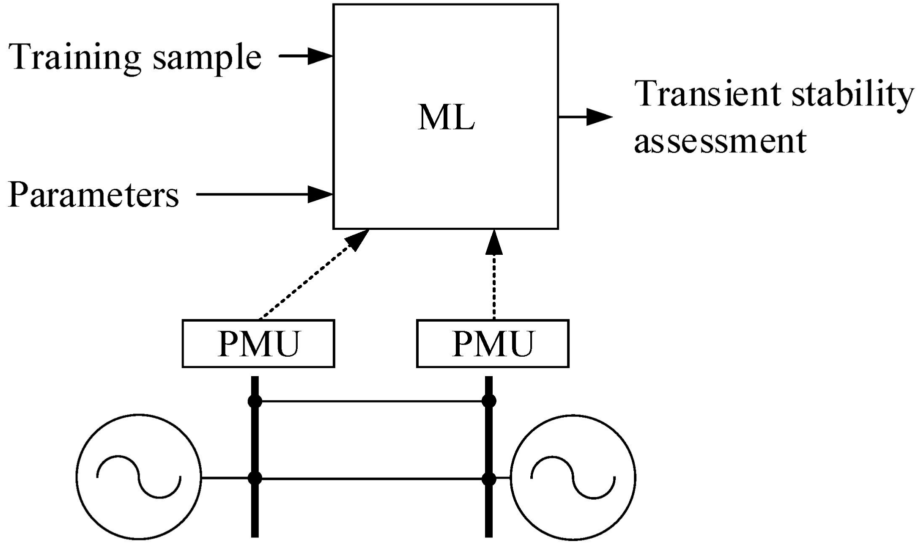
Stability Assessment

- *Stability*: the ability of a smart grid in staying at an equilibrium in case of disturbances
  - or quickly reaching a new equilibrium after a perturbation
- This assessment is done thanks to Phasor Measurement Units (PMU)
  - sensors measuring magnitude and phase angle of voltage and current in different points of the smart grid
- Crucial for ***prevention***:
  - stability assessment supports operators in fixing potential problems related to stability before some external event may make the smart grid unstable



Types of Stability Assessment

- ***Transient Stability Assessment:***
  - determining whether the smart grid components remains synchronised after a huge perturbation (i.e., frequency and phase are aligned). Important for short-term maintenance (e.g., events like lightning strikes)
- ***Small-Signal Stability Assessment:***
  - determining whether the smart grid maintains synchronism when it is under small disturbances (e.g. continuous fluctuations in demand and supply). Important for long-term maintenance
- ***Frequency Stability Assessment:***
  - ensures that the smart grid maintains a steady range of frequencies after a severe system upset or perturbation that results in an imbalance between generation and load
- ***Voltage Stability Assessment:***
  - determining voltage stability to prevent voltage collapse (e.g., sudden load increase/decrease)

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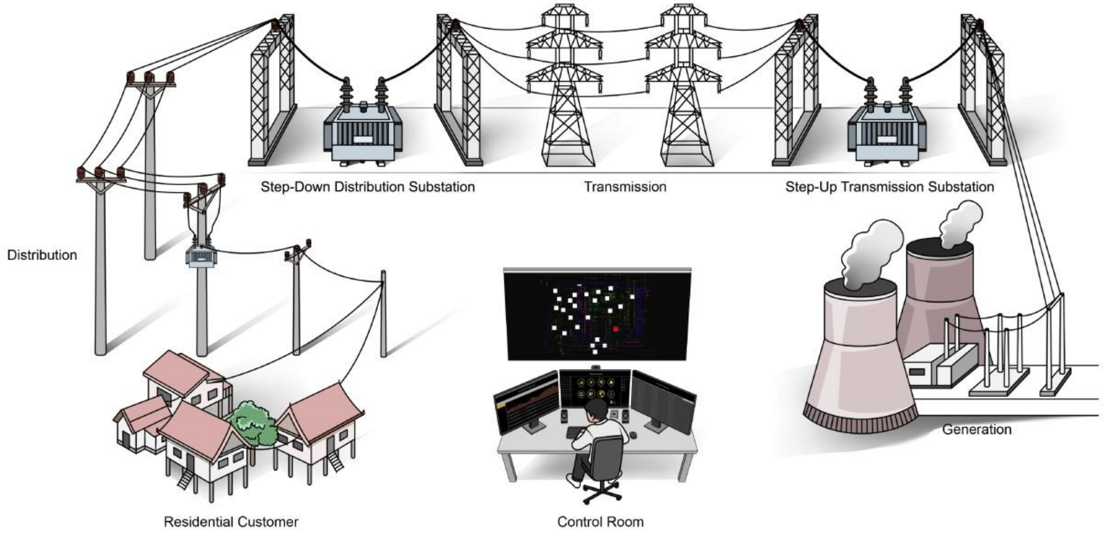
Faults Detection

- Identifying abnormal conditions that can impact the performance of the smart grid
- Crucial for ensuring safety and reliability!
- Managers can rely on fault detection to prevent or minimize disruptions



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Types of faults

- ***Physical devices/components fault:***
  - A physical element does not function properly.
- ***Communication fault:***
  - faults in communication devices/channels
    - e.g., faults in physical links, or failure establishing communications in a critical time period.
- ***Software/Hardware-level fault***:
  - These faults occur when a component of the control center fail to with a command and/or operation.

Smart Energy Management in Smart Homes

Smart Meters

- A smart meter collects information about whole energy consumption in the home
- A fundamental component for ***smart grids***
    - Bi-directional communication
- With respect to traditional meters, they also collect energy information more frequently
- Using smart meters is also referred to ***Non-Intrusive Load Monitoring (NILM)***
    - a single device monitoring the whole home's consumption
    - however, it is not able to provide fine-grained information about each appliance usage

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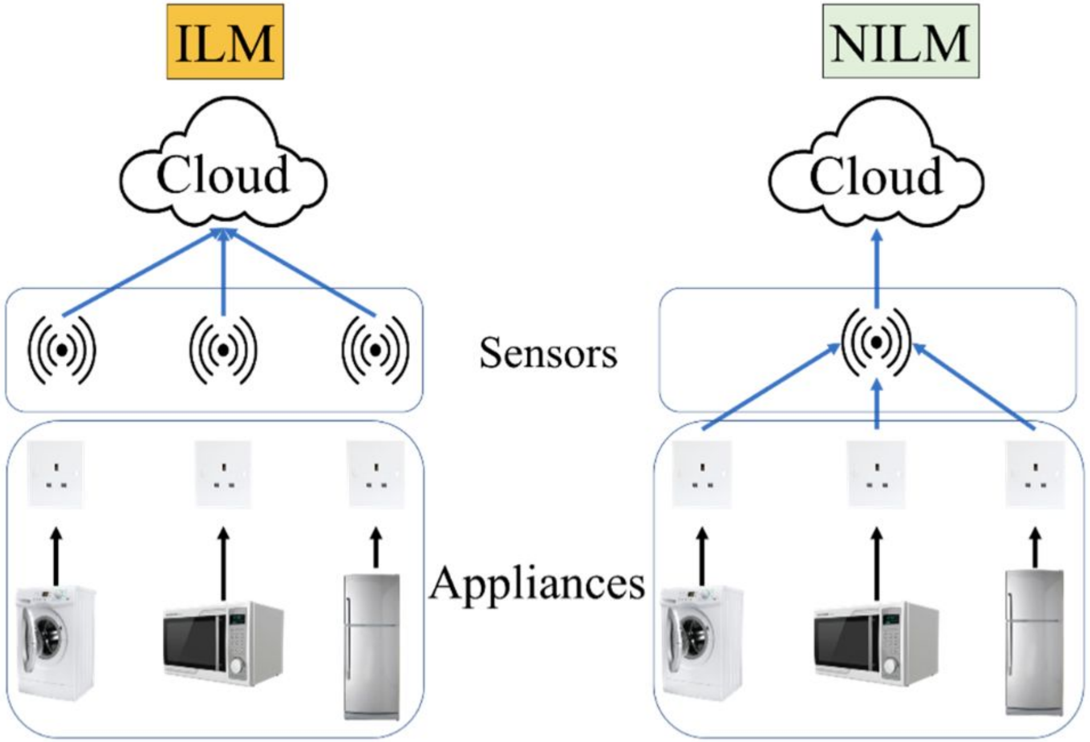
Smart Plugs

- A sensing device that monitors the energy consumption of a specific power plug in the home
- The sensed power consumption is usually transmitted to a home IoT gateway
  - usually through the WiFi network
- Using Smart Plugs is also referred as ***Intrusive Load Monitoring (ILM)***
  - smart plugs can be used to monitor energy consumption at a very fine grained (e.g., for each appliance), but this is more intrusive and costly!

Intrusive vs Non-Intrusive Load Monitoring



Hadi, M.U.; Suhaimi, N.H.N.; Basit, A. Non-Intrusive Load Monitoring and Network. Encyclopedia. https://encyclopedia.pub/entry/25802

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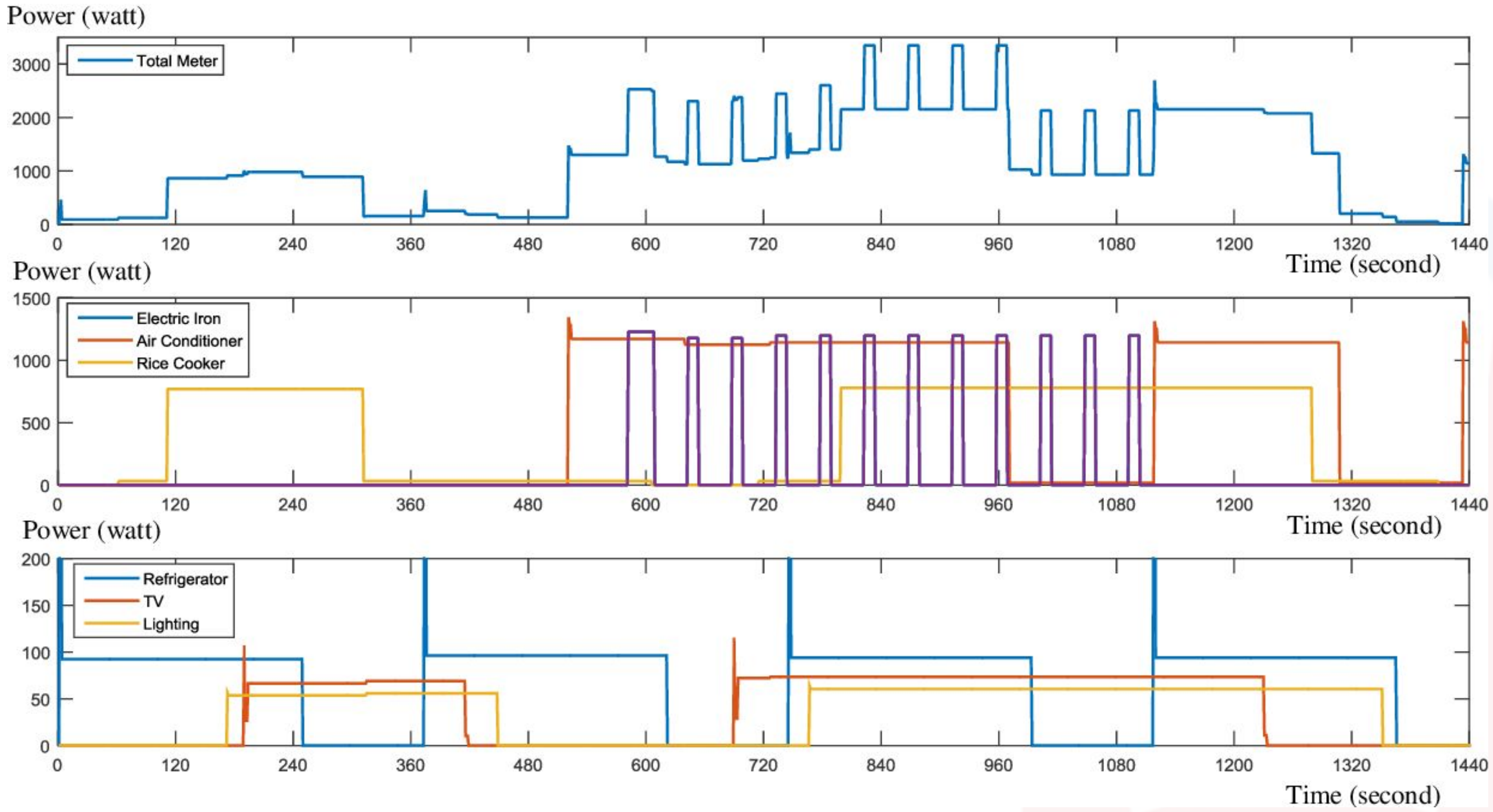
NILM vs ILM: differences in time series

NILM

ILM



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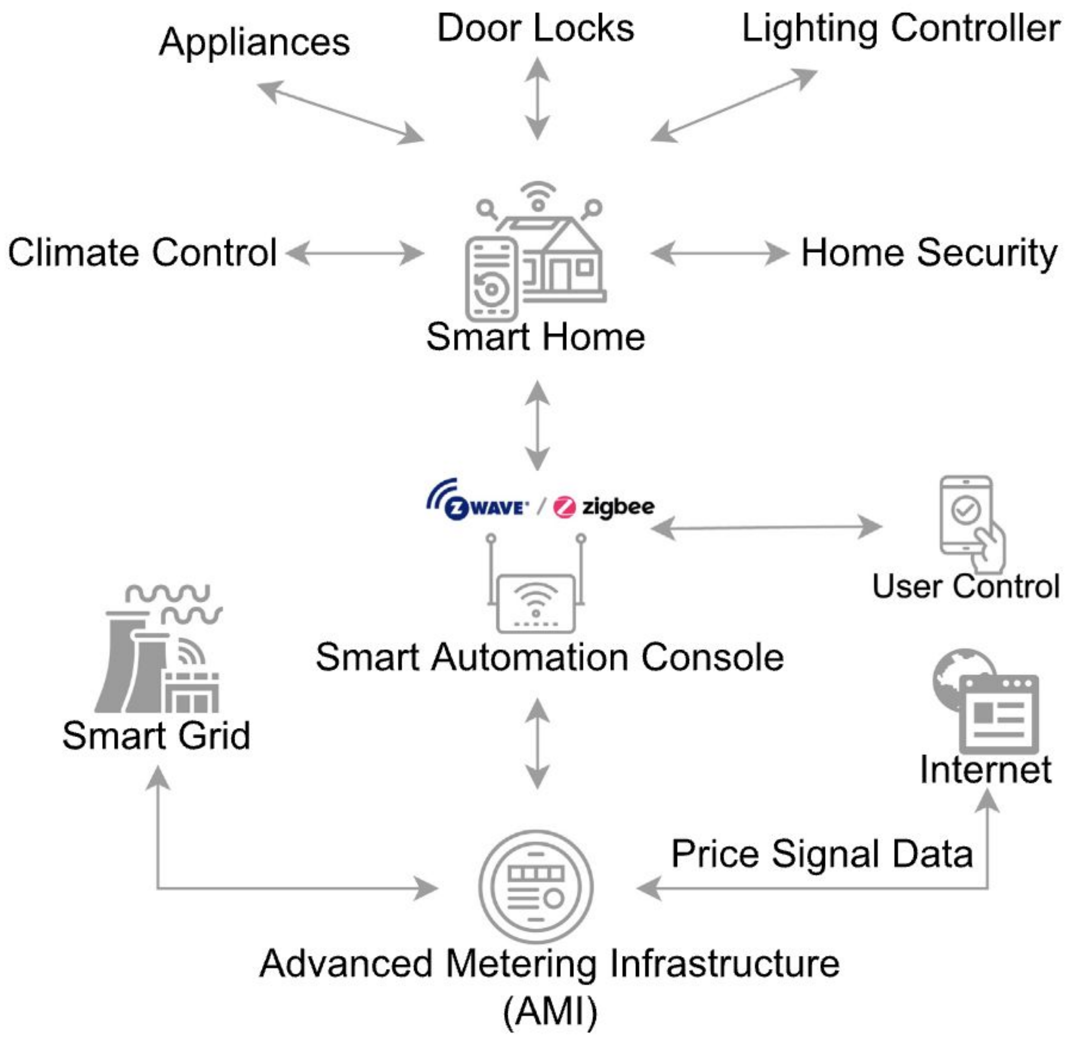
Smart Home Energy Management System



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Smart-homes and energy consumption

- In a smart grid (e.g., a city), the energy loads from the homes significantly impact the overall energy requirements
- Several context variables have an impact, including:
  - types of appliances being used
  - user’s habits
  - external weather conditions
- Monitoring the load in smart-homes can be used for reducing energy consumption
  - e.g., shifting load to off-peak hours or providing energy saving recommendations to the user

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Energy conservation strategies in smart-homes

There are two main strategies (that may be combined) for energy conservation in smart-homes:

- ***User-oriented:*** recognizing habits of the residents, their choices, waste energy patterns
  - improving these aspects is crucial to reduce energy load
- ***Appliance-oriented:*** monitoring the consumption of appliances
  - to automatically schedule them (or to provide guidance to the residents) to reduce energy load

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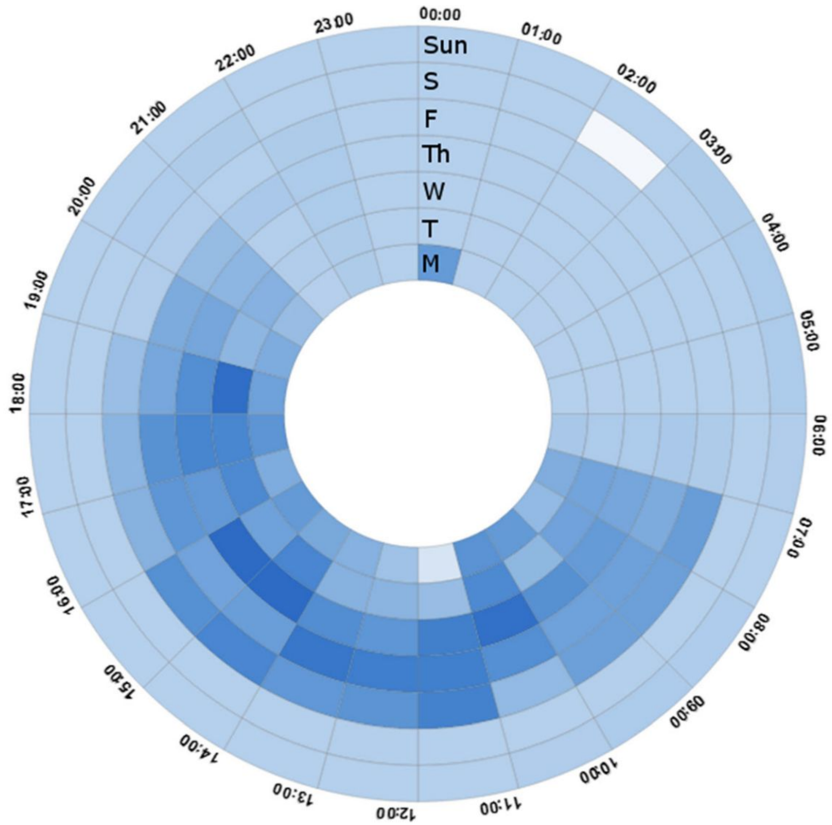
Time and weather

- Seasonal variations have a strong impact on energy consumption
  - e.g., in the summer there is a high use of air conditioning
  - the same holds for weather conditions (e.g., sunny day vs rainy day)
- Time and day are also crucial parameters
- Smart energy management algorithms should take such context information into account!



[Rodríguez Fernández, M., Cortés García, A., González Alonso, I., & Zalama Casanova, E. (2016). Using the Big Data generated by the Smart Home to improve energy efficiency management. *Energy Efficiency*, *9*, 249-260.]

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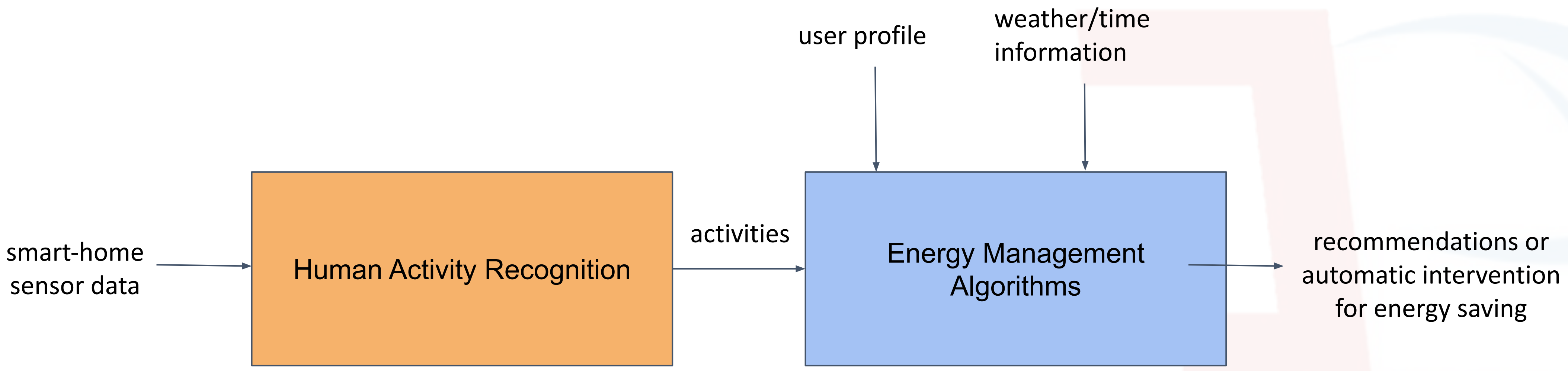
Understanding user habits

- Indoor Localization and Human Activity Recognition methods can be used to derive the daily habits of the residents
  - ***personalization*** is crucial, since different persons have different habits

- The user profile can also be depicted by the usage of the appliances (e.g., when turned ON/OFF)

- Habits can then be matched with energy patterns!

Activity Recognition for Energy Management



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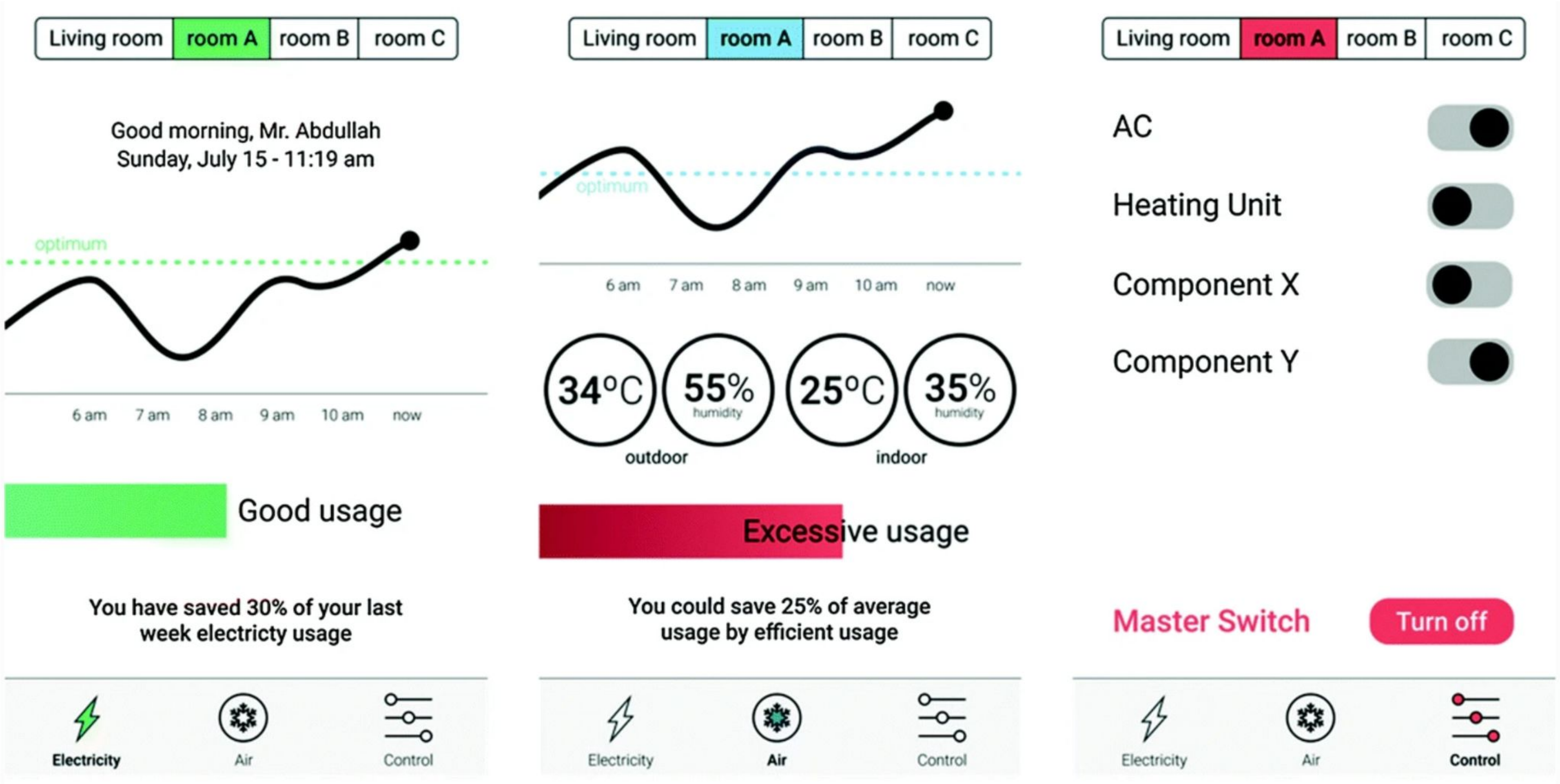
User Recommendation - Visualization



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[Alsalemi, A., Bensaali, F., Amira, A., Fetais, N., Sardianos, C., & Varlamis, I. (2020). Smart energy usage and visualization based on micro-moments. In *Intelligent Systems and Applications: Proceedings of the 2019 Intelligent Systems Conference (IntelliSys) Volume 2* (pp. 557-566). Springer International Publishing.]

Load Forecasting in Smart Homes

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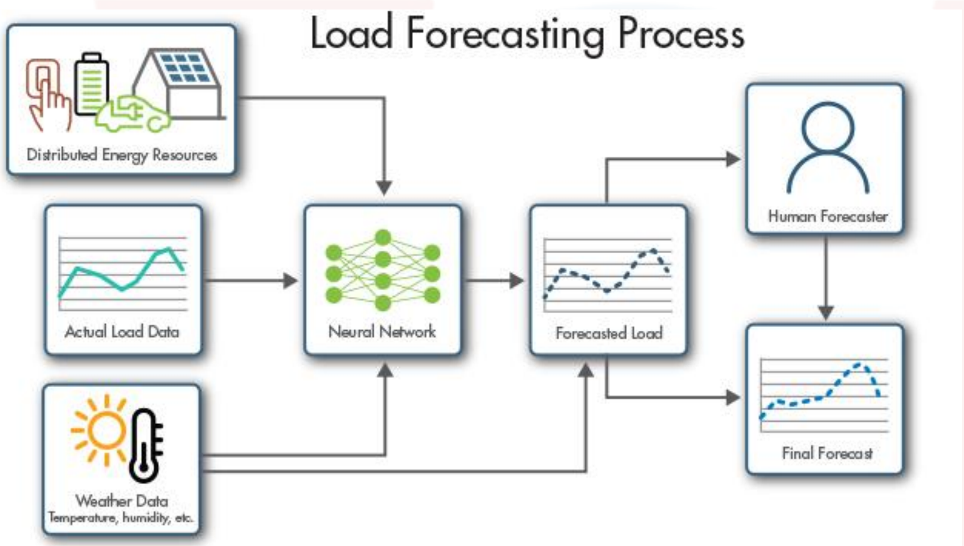
Load Forecasting in Smart Homes

- One of the main methods to reduce energy consumption is to predict future load in the home
  - By using historical energy data as well as context information like the user's habits, availability of renewable sources, etcetera
  - Accurate weather forecast may also provide valuable information
- Load forecasting can be done at different granularities (e.g., for the whole home or appliance-level)
- Short-term forecasting is the most common:
  - to enable dynamic pricing and reducing peak demand
  - to better perform scheduling of appliance usage to save from electricity bills



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Load Forecasting "ingredients"

Setting

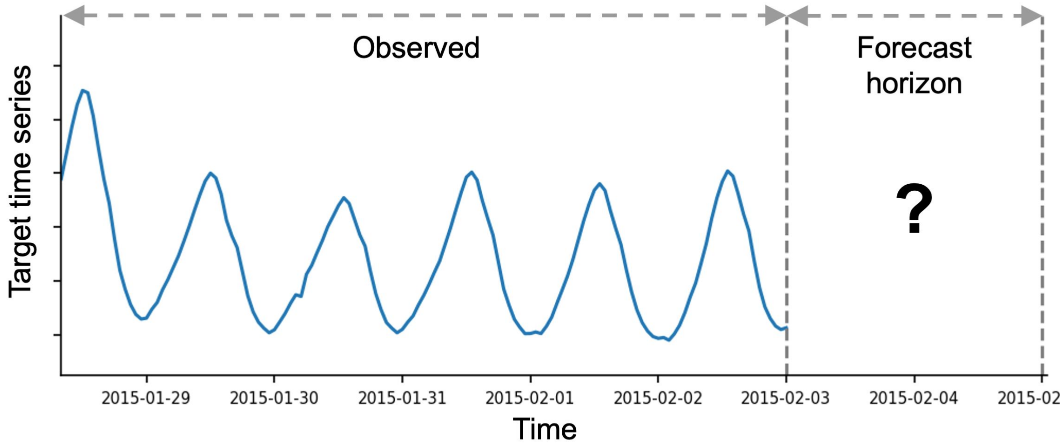
- A smart home contains multiple electrical appliances.
- Power consumption measured at regular time intervals (e.g., minutes, 15 min, hours).

Observation Horizon

- Time series of historical data that is available to the model.
- Defines how much past data is used as input.

Forecast Horizon

- Time window over which future consumption must be predicted.
- Can range from minutes to hours or days ahead.



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Load Forecasting goal

- The model takes as input:
  - Past consumption measurements defined by observation horizon
  - Optional contextual variables (time of day, weather, occupancy patterns)
- The goal is to predict the energy values in the forecast horizon time window
- Forecasting can be done at different granularity levels:
  - Appliance level
  - Device category level (e.g., HVAC, lighting)
  - Whole-home consumption

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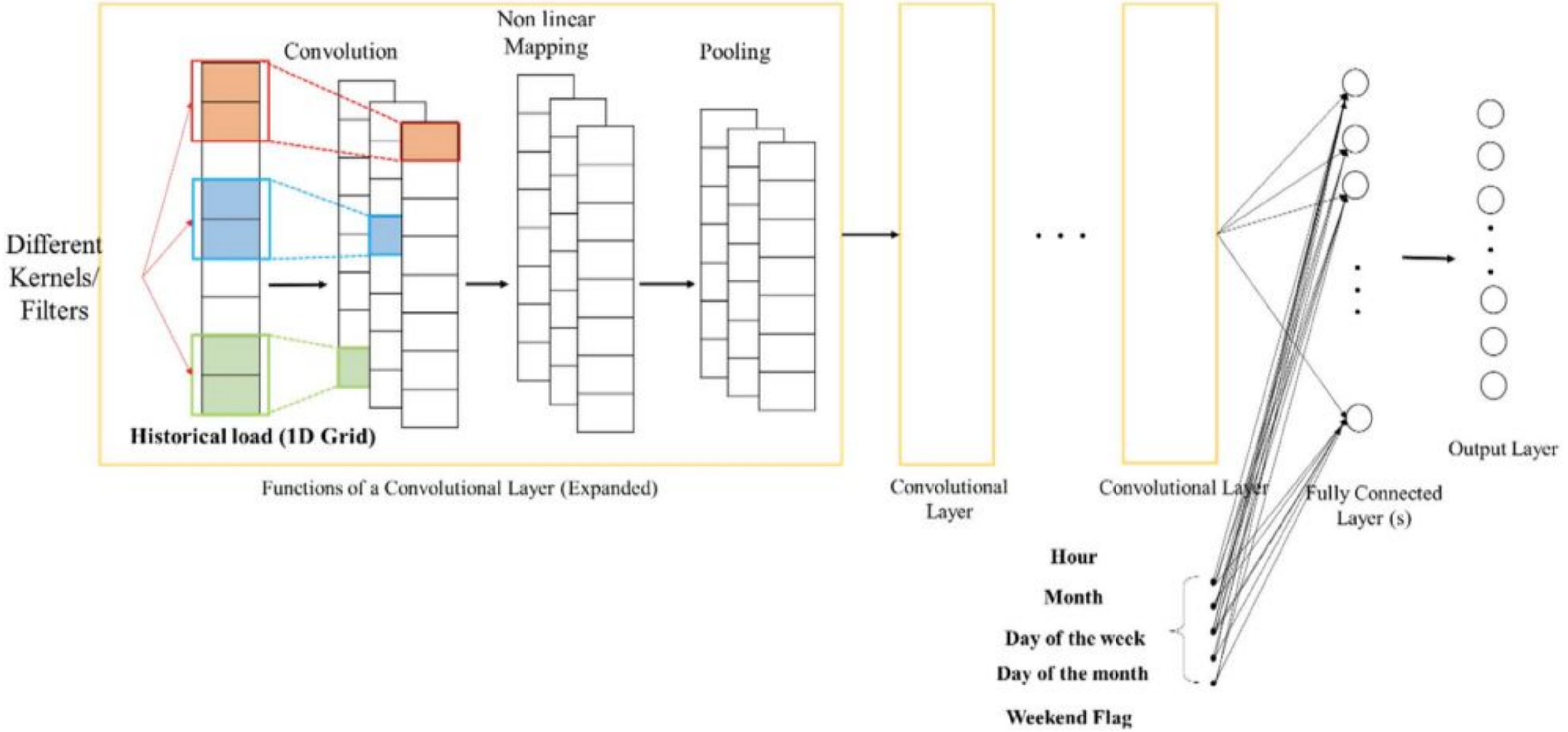
Load Forecasting: a CNN-based approach

- 1D convolutional layers to extract features from historical data
- In the fully connected layers, context features (e.g., hour, month) are introduced!
- Output: a fixed number of future energy values (regression)



[Amarasinghe, K., Marino, D. L., & Manic, M. (2017, June). Deep neural networks for energy load forecasting. In 2017 IEEE 26th international symposium on industrial electronics (ISIE) (pp. 1483-1488). IEEE.]

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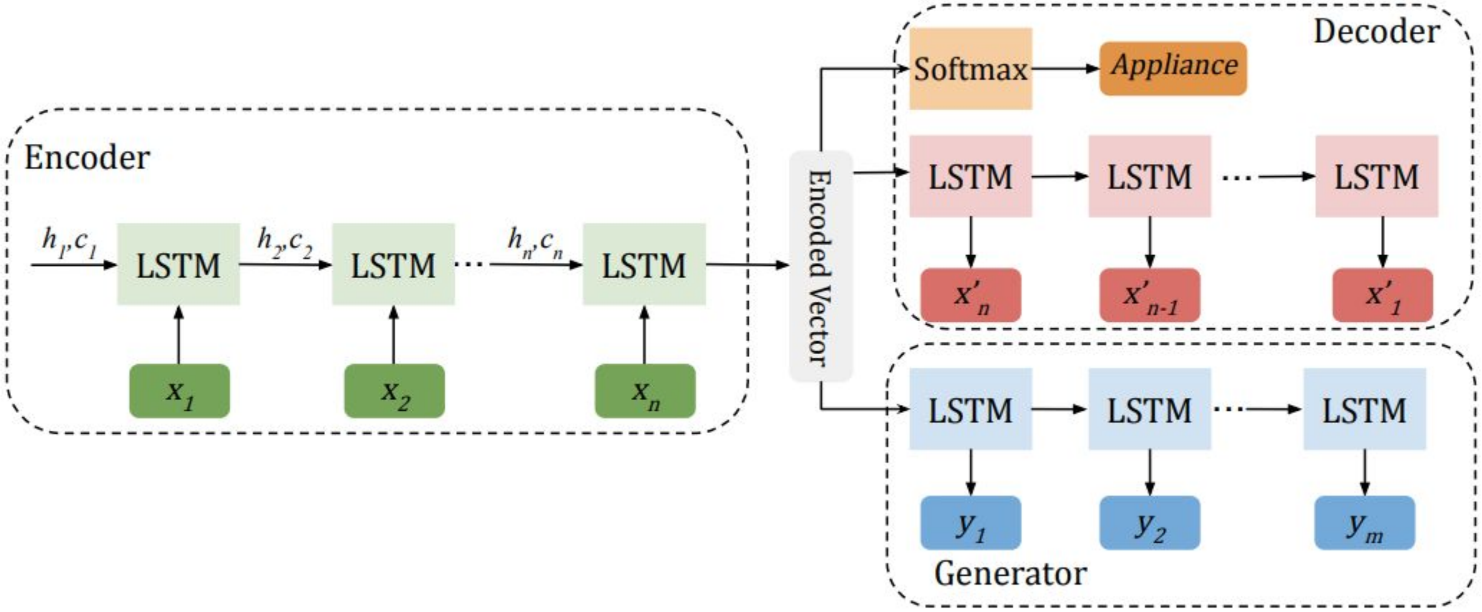
Load Forecasting: a seq2seq approach

- **Encoder**: it learns a latent representation of past energy data
- **Decoder:** it learns temporal dependencies by reconstructing the input sequence in the reverse form (giving more weight the most recent observations)
    - optional: it may also reconstruct the appliance originating the signal
- **Generator:** it predicts the next *m* data points in the sequence



M. Razghandi, H. Zhou, M. Erol-Kantarci and D. Turgut, "Short-Term Load Forecasting for Smart Home Appliances with Sequence to Sequence Learning," *ICC 2021 - IEEE International Conference on Communications*, Montreal, QC, Canada, 2021, pp. 1-6.

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Appliances Scheduling based on Reinforcement Learning

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Appliance scheduling

- Short-Term Load Forecasting can be used to schedule appliances in low-price energy hours based on consumption
- However, a challenge is to understand whether it is possible to actually defer appliance usage and when!
- Idea:
  - identify appliances consuming extra energy
  - predict when their usage would minimize costs
  - schedule them by minimizing user's discomfort
    - we will see an approach based on Reinforcement Learning!

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Categories of Home Appliances

- ***Non-deferrable:*** cannot be shifted or scheduled (e.g., refrigerator, TV)
- ***Deferrable:*** can be scheduled (or shifted) based on energy requirements. When turned ON, these appliances can NOT be halted and shifting their schedule may impact discomfort level (e.g., washing machine). Preferable in off-peak times.
- ***Controllable:*** can be operated at different power levels. This may also depend on environment temperature (e.g., air-conditioner). Changing power level may impact discomfort level.

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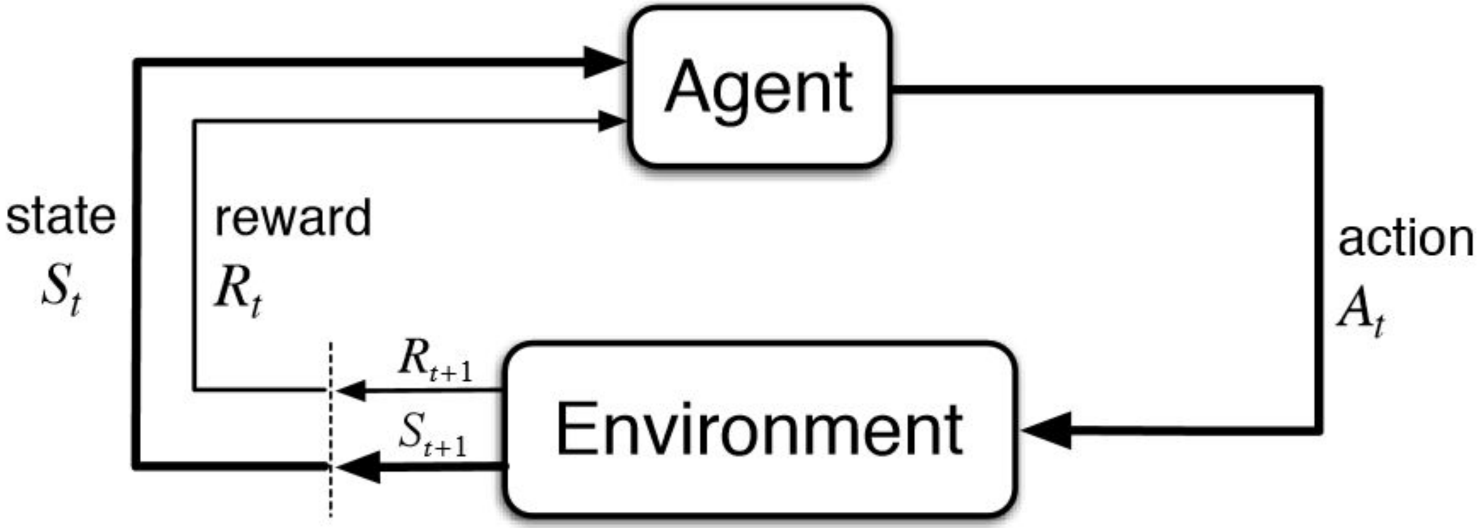
Reinforcement Learning

- Problems involving an ***agent*** interacting with an ***environment***, which provides ***numeric reward signals***

- Goal: Learn how to take actions on the environment in order to maximize reward



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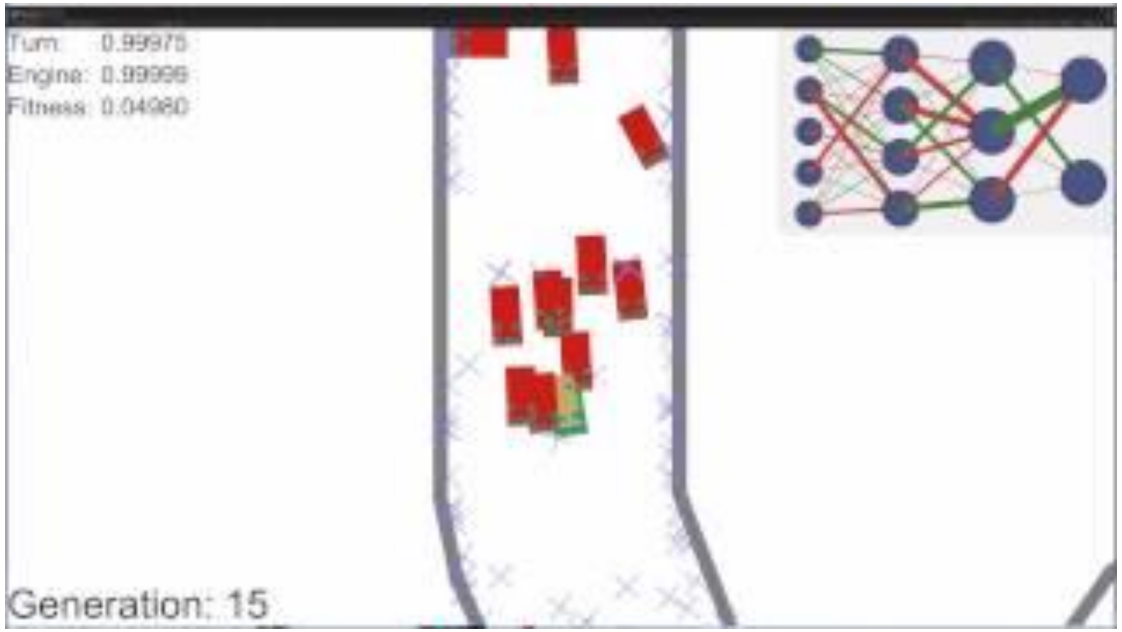
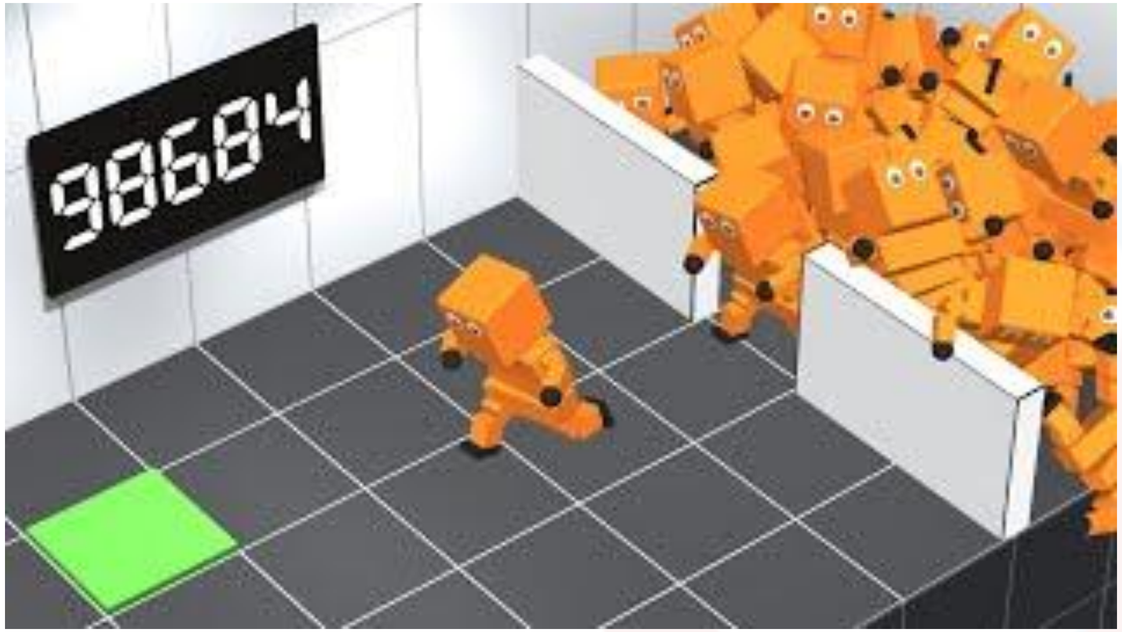
Reinforcement Learning in Practice





“Classic” Reinforcement Learning

- The goal is estimating a function *Q(s,a)*
  - given an observed state *s*, it estimates the potential reward in taking an action *a*
  - the objective is choosing the action maximizing the expected reward (unknown before taking the action) based on the current state
  - after taking the action, the reward is observed in the environment and the estimate of *Q(s,a)* is updated

$$\underbrace{Q(S_t, A_t)}_{\text{New Q-value estimation}} \leftarrow \underbrace{Q(S_t, A_t)}_{\text{Former Q-value estimation}} + \underbrace{\alpha}_{\text{Learning Rate}}[\underbrace{R_{t+1}}_{\text{Immediate Reward}} + \underbrace{\gamma max_a Q(S_{t+1}, a)}_{\text{Discounted Estimate optimal Q-value of next state}} - \underbrace{Q(S_t, A_t)}_{\text{Former Q-value estimation}}]$$

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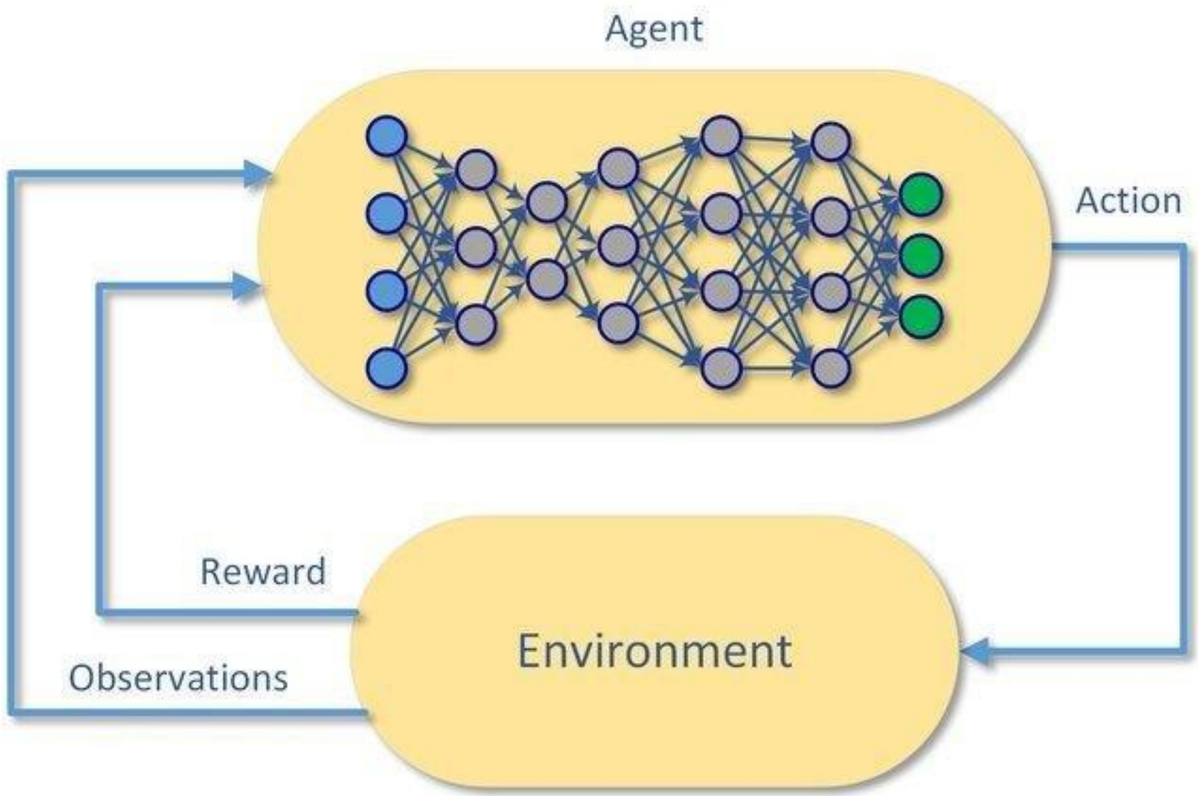
Deep Reinforcement Learning

A recent evolution of Reinforcement Learning where the Agent learns to take actions on the environment by using deep learning.

Basically, the neural network implicitly learns the Q(s,a) function by maximizing the reward collected in the environment.



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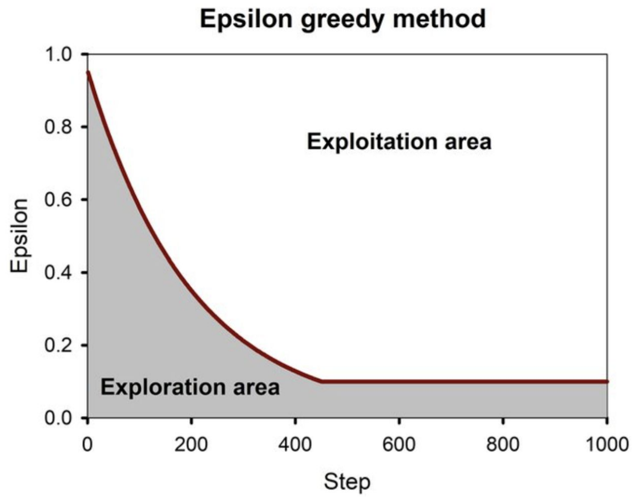
Exploration VS Exploitation

- At the very begin, the agent needs to explore the possible actions by choosing them at random (the policy is not known): ***exploration***
- After some time, the agent learns a policy and the best action can be chosen: ***exploitation***
- There is the need to balance exploration and exploitation to avoid reaching local optima
- Usually the adopted approach is ***ε-greedy***:
  - the probability of **exploiting** is 1-ε, while the probability of **exploring** is ε
  - ε is high at the begin, and it decays as the learning process progresses



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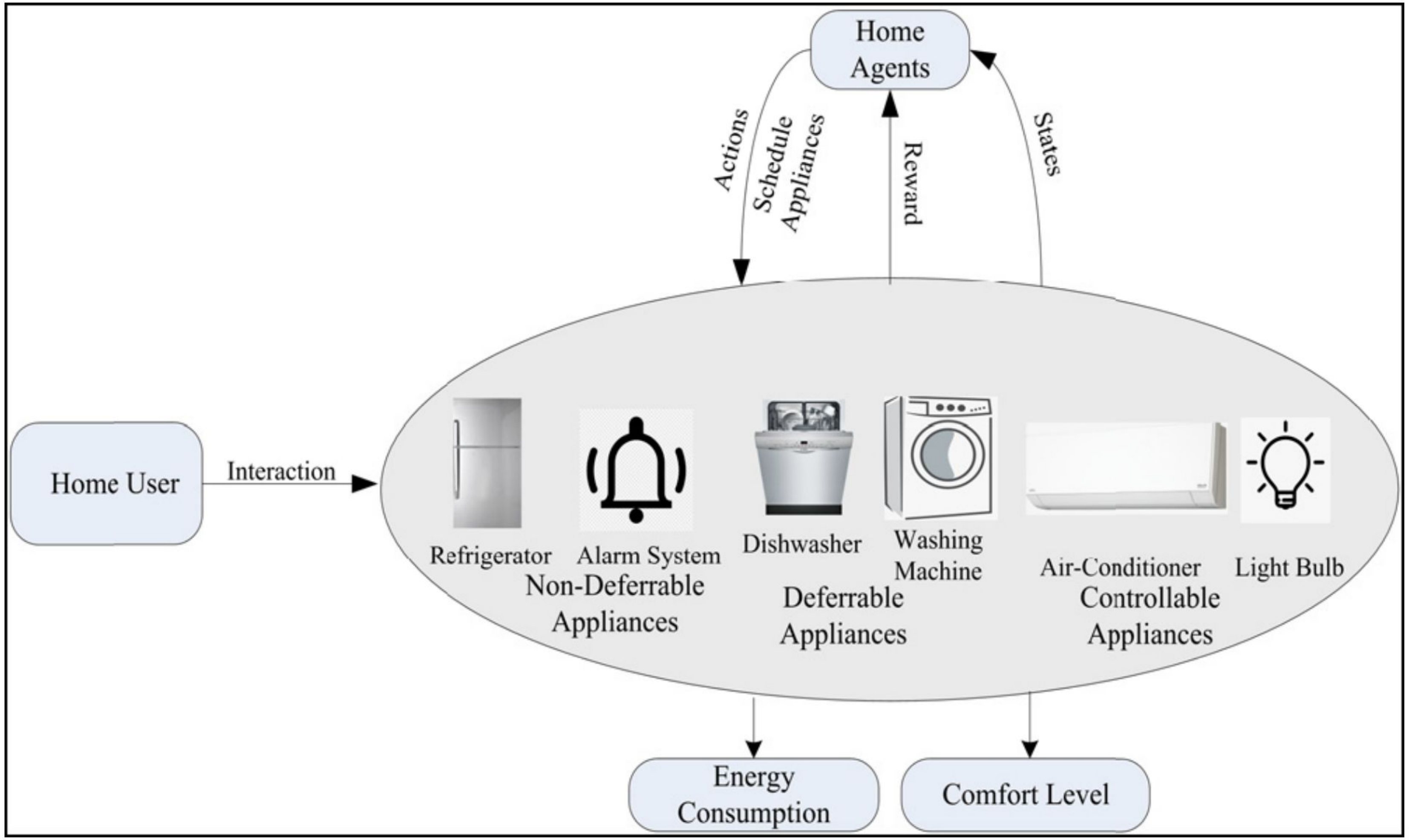
Reinforcement Learning in Energy Management

- **Agent:** the energy management system
- **Environment:** observations of energy consumption (e.g., appliance level), data from other sensors, time, weather, user activities, user's comfort level, etcetera
- **Actions:** energy saving strategies
  - e.g., switch off appliances with low priority, change power level, defer appliances usage,...

Reinforcement Learning for Energy Management



[Khan, M., Seo, J., & Kim, D. (2020). Real-time scheduling of operational time for smart home appliances based on reinforcement learning. *IEEE Access*, *8*, 116520-116534.]

How to train the model?

- The energy management system needs to learn a ***policy***: which action to take based on the current state of the home?
  - optimization criteria: maximize comfort level and minimize energy consumption
- A ***reward*** is automatically computed by analyzing the impact of performing an action to the environment
  - e.g., positive values when the action is good in terms of comfort and energy consumption, negative value otherwise
  - it is also possible that an action has no impact on the environment, in such cases the reward is 0
- The goal of the agent is to choose actions that maximize rewards on the long term

Evaluating the User’s Comfort

- Possible ways to estimate the user’s comfort in the reward:
  - using the average waiting time rate (WTR) of the operation of appliances
    - users typically prefer starting the operation of their smart home appliances with short delays
    - hence minimizing WTR increases the comfort
  - considering environmental conditions (e.g., indoor temperature, humidity, air quality) that may be affected by appliances
  - using emotion/stress recognition tools

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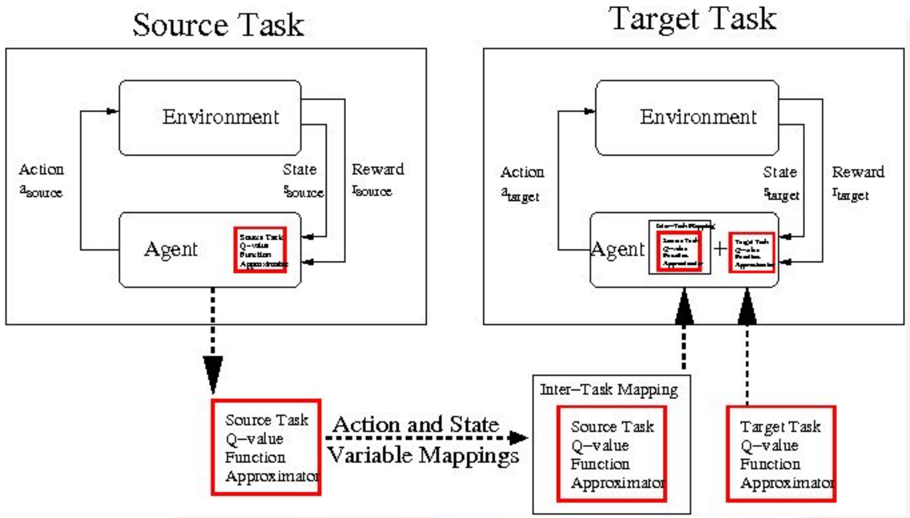
Exploration VS Exploitation

- As previously mentioned, the agent has to balance exploration and exploitation to find a good energy management policy
- However, this is a problem in smart-homes, since random actions may negatively impact the user's comfort
    - to mitigate this problem, the action decision module may be pre-trained in a controlled environment and then fine-tuned in the target home with ***transfer learning***



The Load Disaggregation Problem

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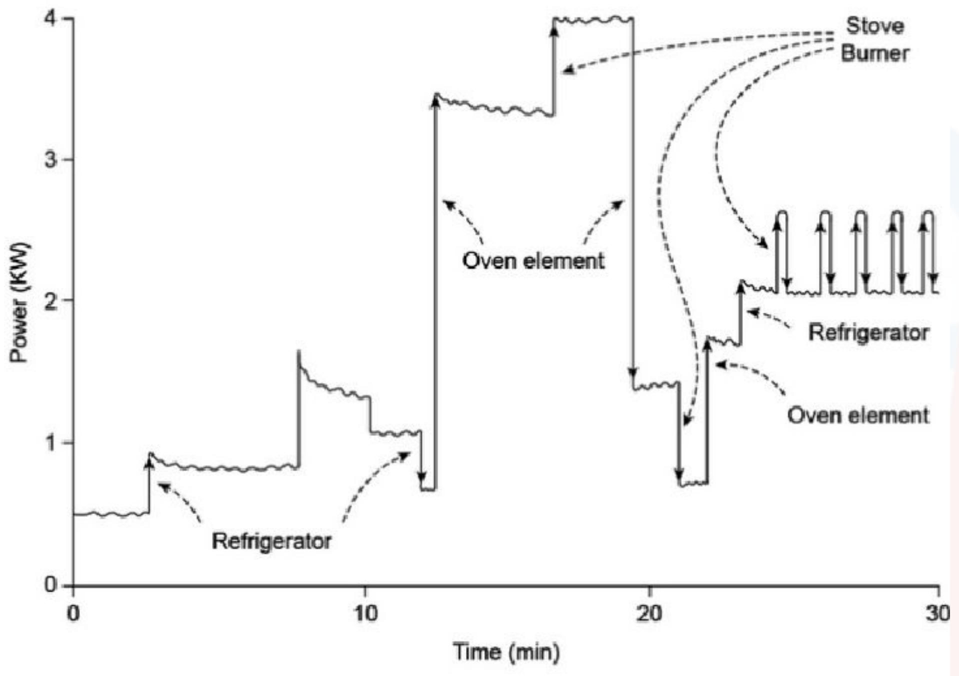
Load Disaggregation (Non-Intrusive Load Monitoring)

- Consider a smart-home where only a single smart meter collects energy consumption data of the whole home
  - or a single smart plug is connected to many appliances
- Is it possible to understand how each single appliance contributes to the overall energy consumption?
  - this problem is referred as ***load disaggregation*** or ***non-intrusive load monitoring (NILM)***



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Total aggregated load

$$P(t) = \sum_{i=1}^{n} P_i + e(t)$$

total load
at time t

load of the i-th
appliance

a noise error term

Goal of load disaggregation: observing only P(t) and estimating each P_i

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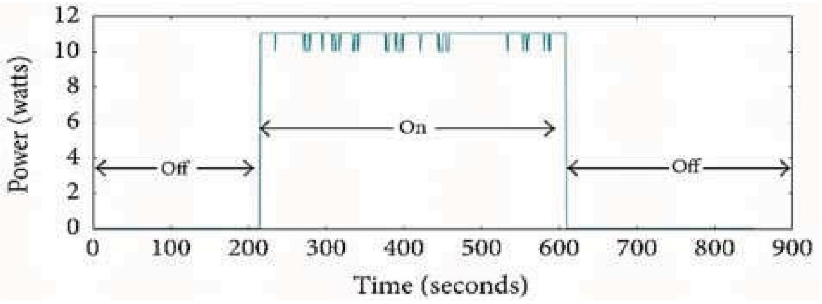
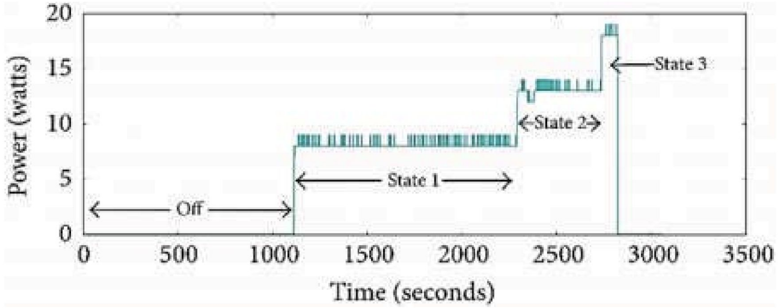
Load patterns depend on the appliance

- Each appliance is likely to have a characteristic load signature

- Type 1: On-Off (or Single-State) appliances
    - they only have two states of operation
    - examples: table lamp, toaster
    - the ***easiest*** to detect

- Type 2: Multi-State appliances
    - more than two states of operation (known transitions)
    - examples: electric fan, washing machine
    - they typically exhibit a repeated pattern





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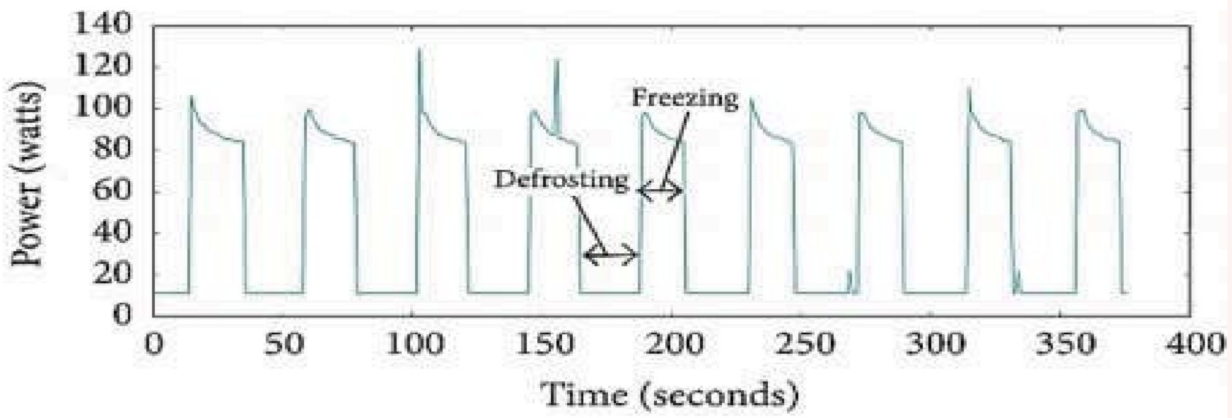
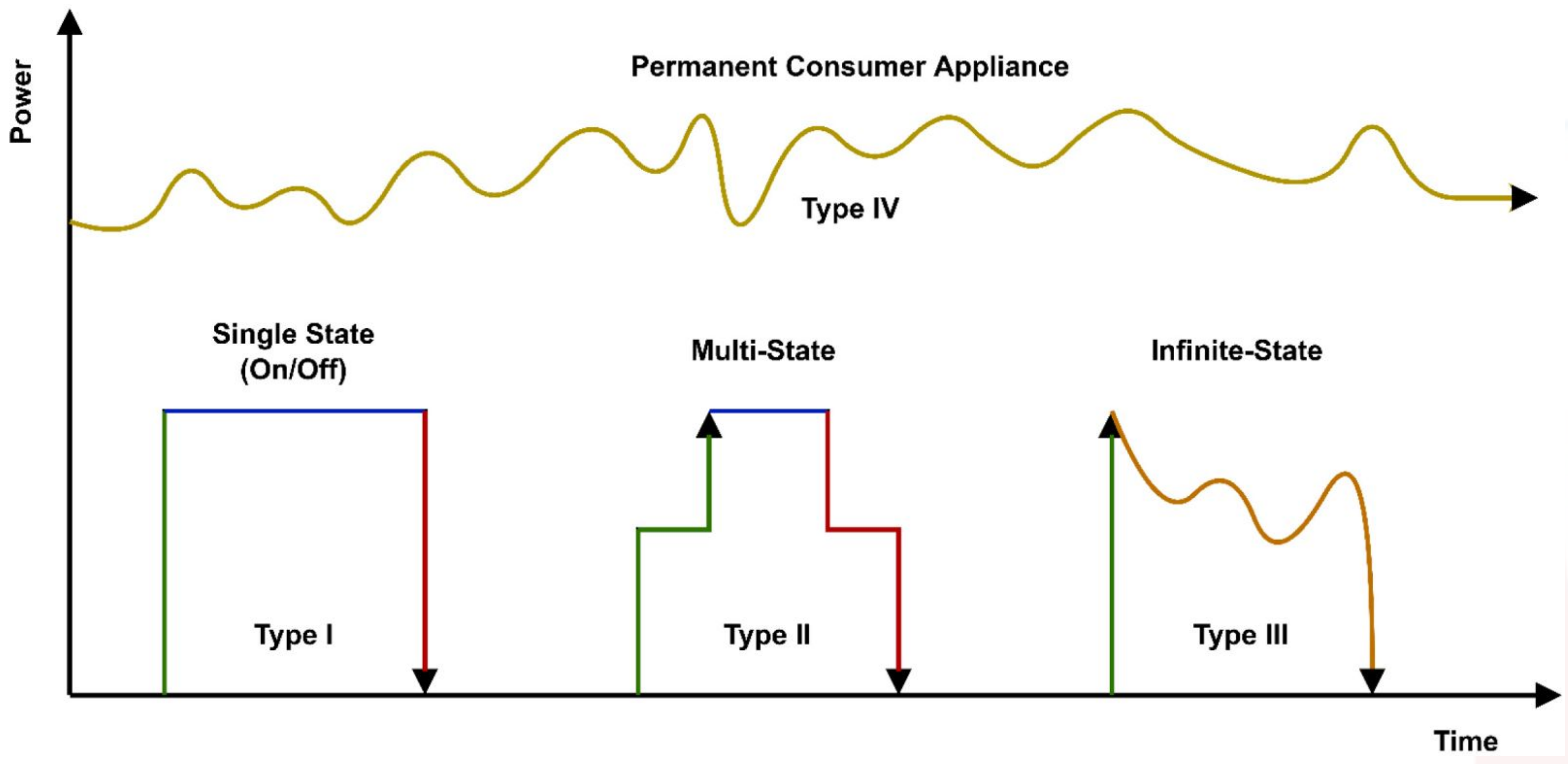
Load patterns (cont.)

- Type 3: Continuously Variable (or Infinite-State) Appliances
  - consumption not fixed and keep changing
  - examples: laptop, electric drill
  - the most challenging ones

- Type 4: Permanent Consumer Appliances
  - remain active and consuming constant power over time periods
  - examples: refrigerator, TV receiver



Load Patterns (summary)



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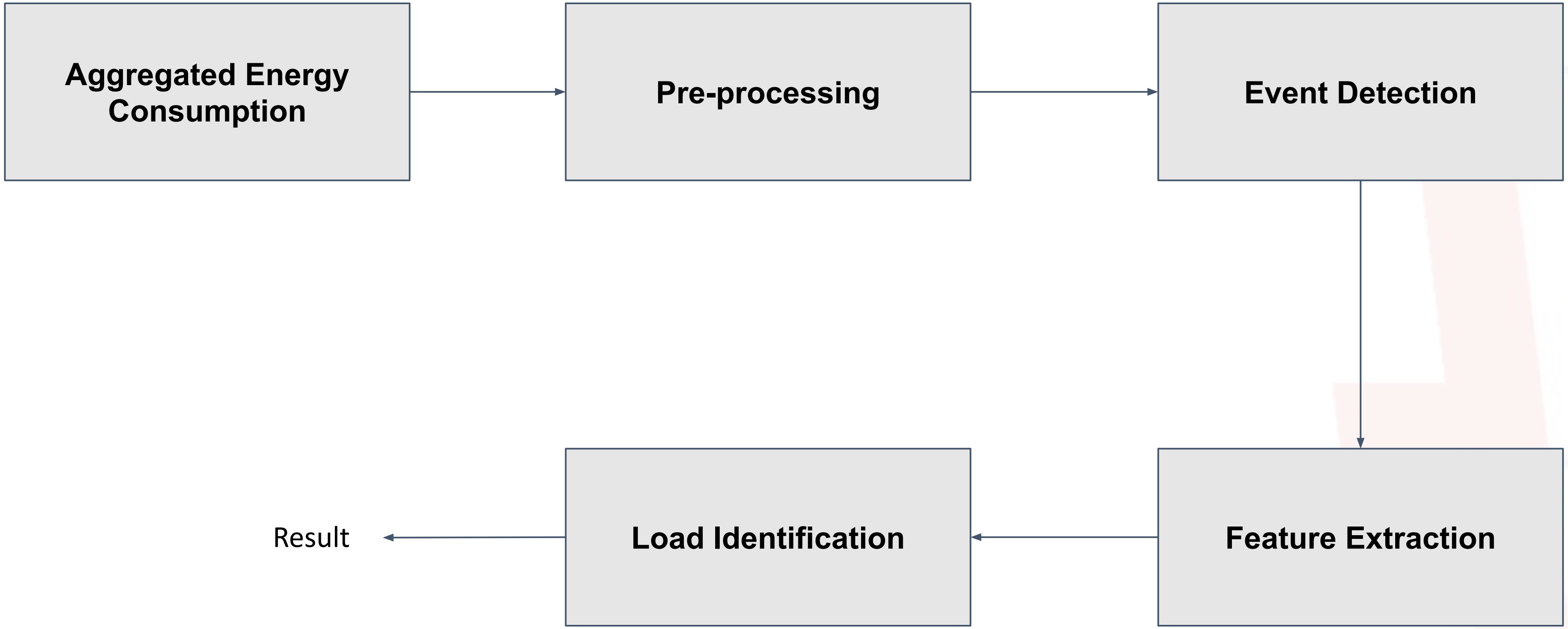
Load Disaggregation “Classic” Pipeline



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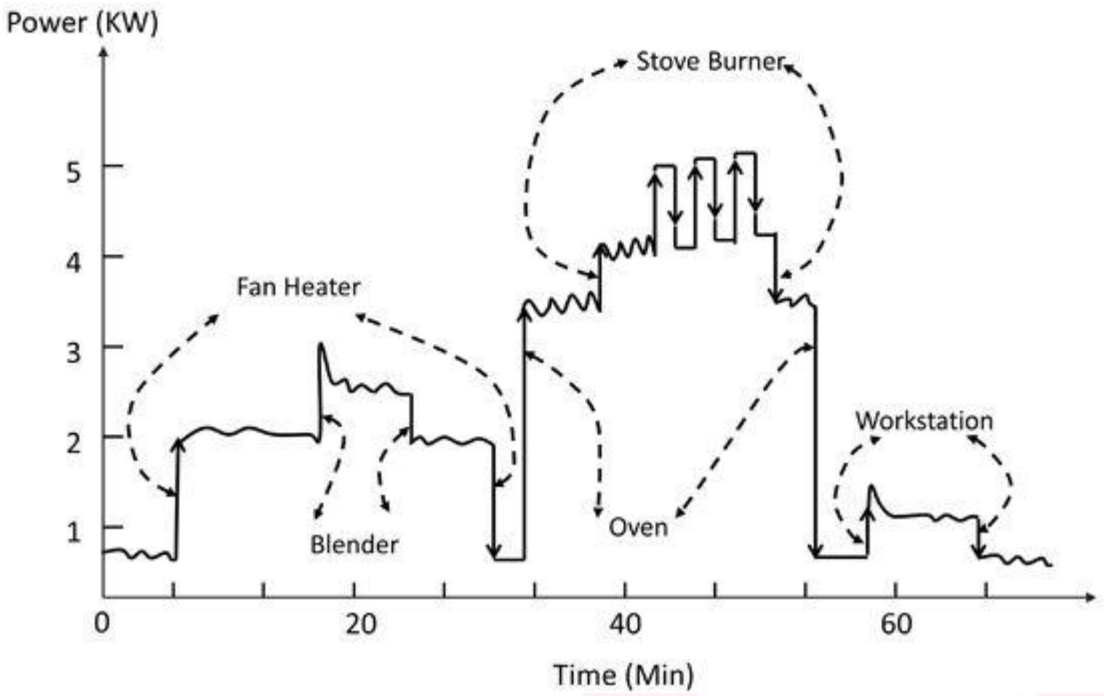
Event Detection

- Goal: segmentation by identifying changes in energy consumption that may be attributed to change of states in the appliance
- Two types of events:
  - ***Steady state changes:***
    - e.g., variations of energy consumption occurring while the appliance is in a “stable state”
  - ***Transient changes:***
    - capturing specific transitions from one state to another



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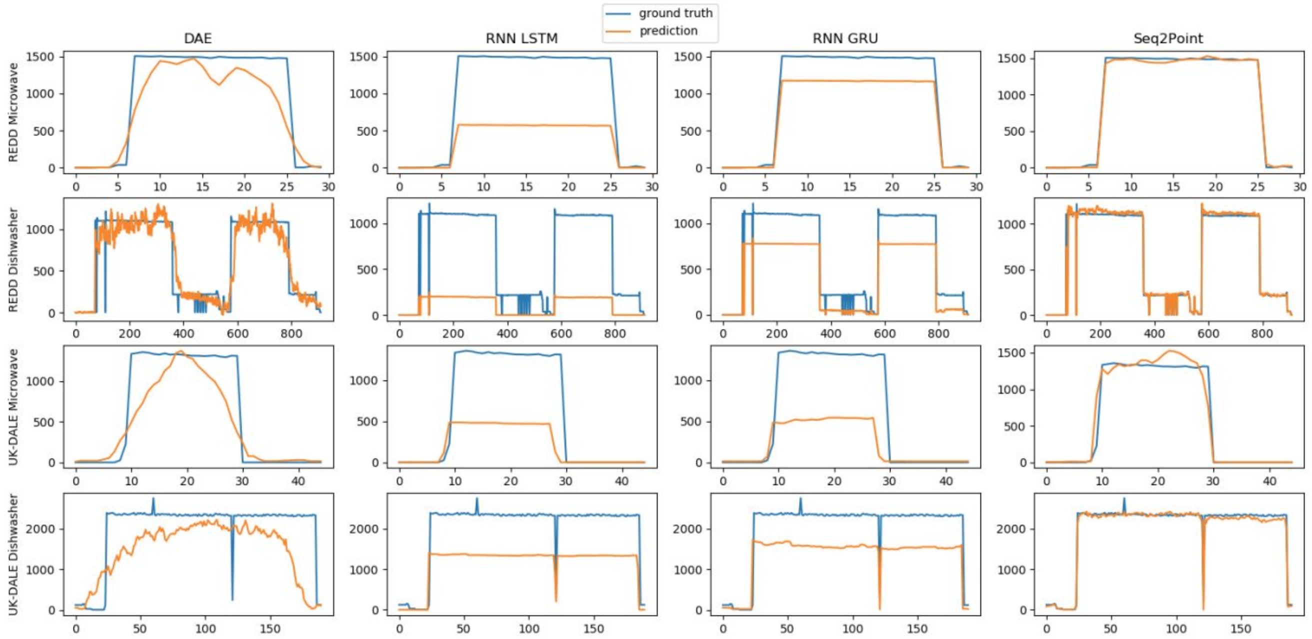
A metric for NILM: MAE

Mean Average Error (MAE)

$$MAE = \frac{1}{T}\sum_{t=1}^{T}\left|\widehat{y_t^{(i)}} - y_t^{(i)}\right|$$

For each appliance *i*, the goal is to minimize the error in estimating the energy consumption!

It is a ***regression*** problem with multiple outputs.



NILM using deep learning

- In NILM, deep learning methods have become the standard solution
- Why?
  - they learn features directly from raw data, removing the event detection problem
  - they learn complex load signatures
  - they learn temporal dependencies between appliances
  - more robust to noise and scalable

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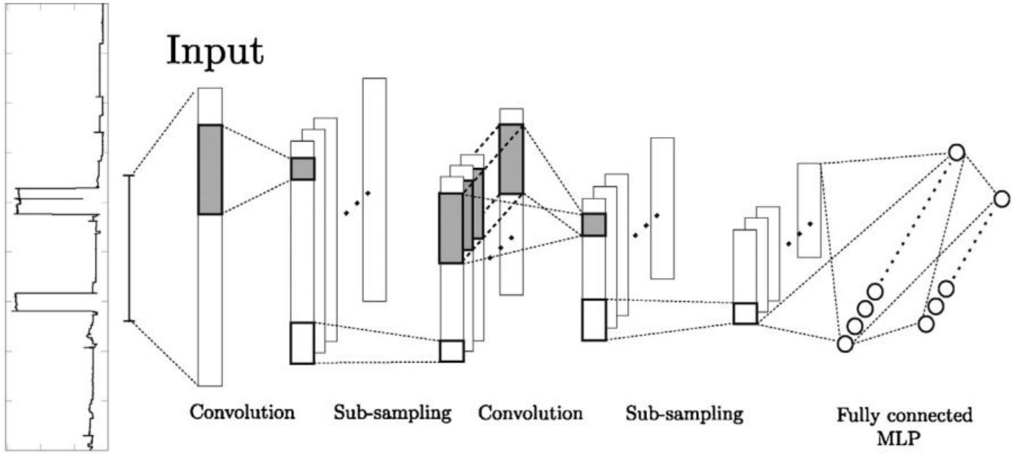
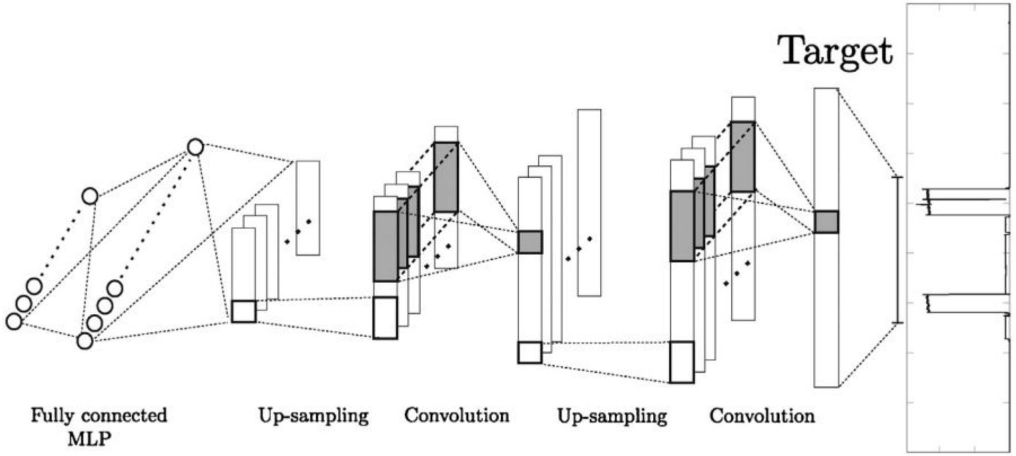
NILM using a Denoising AutoEncoder

- ***Denoising:*** extracting *clean* signal from a noisy one
  - The encoder receives as input the aggregated energy consumption from the smart meter during a temporal window
  - This input is mapped in the latent space
  - The decoder maps the latent representation into power consumption for each appliance

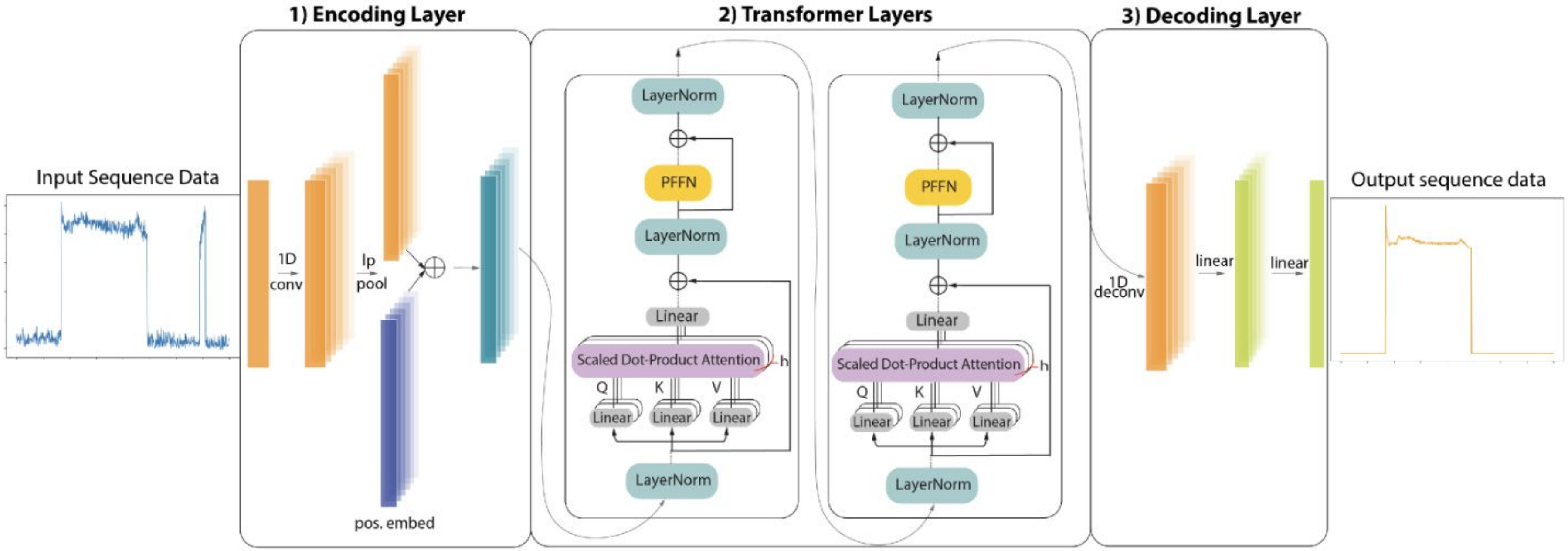


(a) Encoder network. The input signal is the aggregated power consumption.



(b) Decoder network. The target signal is ground truth power consumption of each appliance.

NILM using Transformer Architectures



Sykiotis S, Kaselimi M, Doulamis A, Doulamis N. ELECTRIcity: An Efficient Transformer for Non-Intrusive Load Monitoring. *Sensors*. 2022; 22(8):2926. https://doi.org/10.3390/s22082926

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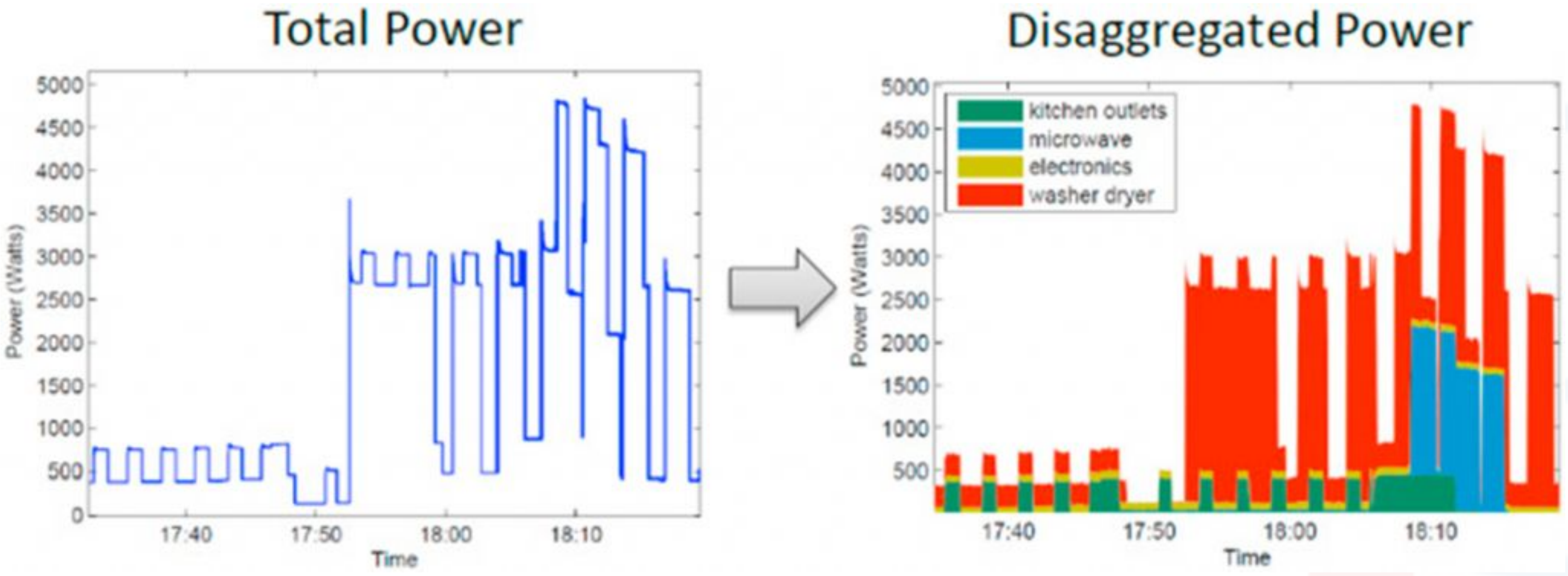
The output



https://www.zyax.se/posts/2018/10/19/Non-Intrusive-Load-Monitoring

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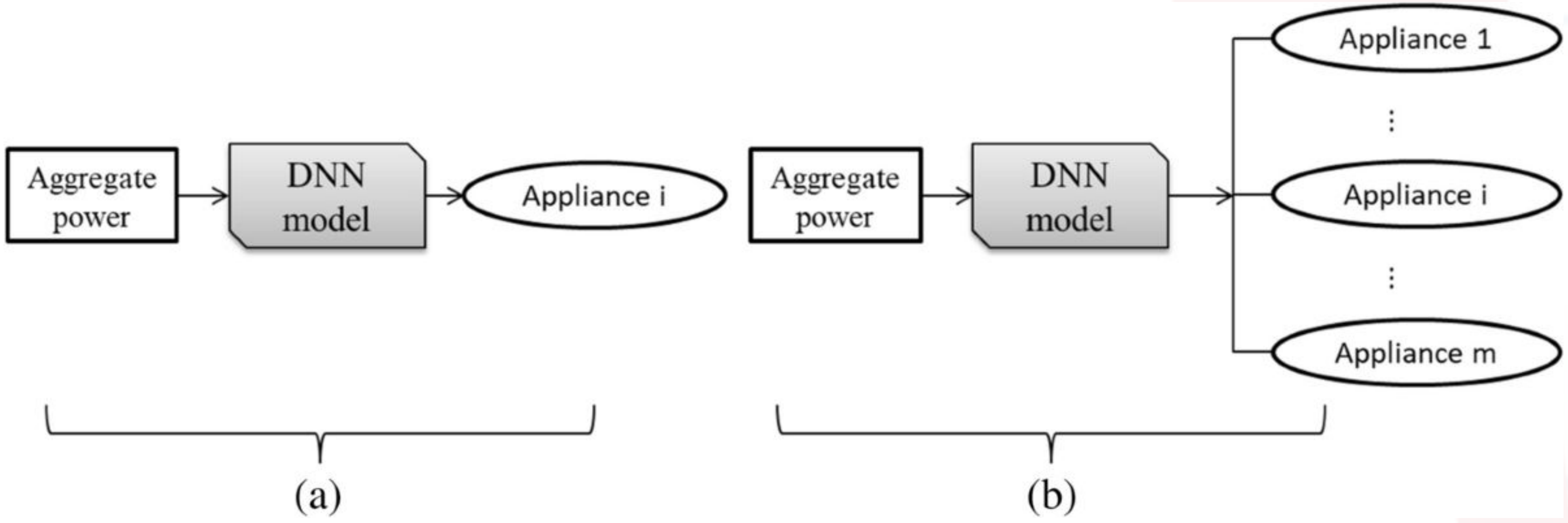
Single-Appliance Models vs Multi-Output Models in Deep Learning NILM

Two approaches:

- 1) Train a separate neural network for each appliance
    - simpler learning task: each model focus on one output appliance
    - high computational costs: a new model must be trained for every appliance!
    - poor scalability
    - no shared learning of common patterns among appliances
- 2) A single neural network takes providing the estimated power for multiple appliances simultaneously.
    - More efficient training and deployment since only one model is required
    - Shared feature extraction across appliances
    - Harder optimization because the model must learn several tasks simultaneously (need more complex models and more data)
    - Adding a new appliance requires modifying the architecture (e.g., adding a new output head and fine-tuning).

NILM: single-appliance vs multi-output models



Single-appliance model

Multi-output model

How to train Load Disaggregation Models?

- Load Disaggregation is usually implemented as a supervised regression task
  - the ground truth is the energy consumed by each appliance
- Labelling is a challenge!
  - each home has different number and type of appliances and it's difficult to have a single model capable of generalizing
  - a possible solution, in an initial phase, is to use a plug sensor for each appliance to collect ground truth (submetering)
    - **drawback:** costly (you need to buy plug sensors) and unrealistic (once you bought them, why getting rid of them?)
  - another solution is to manually turning on/off appliances to annotate the load of each one
    - it may be prohibitive!

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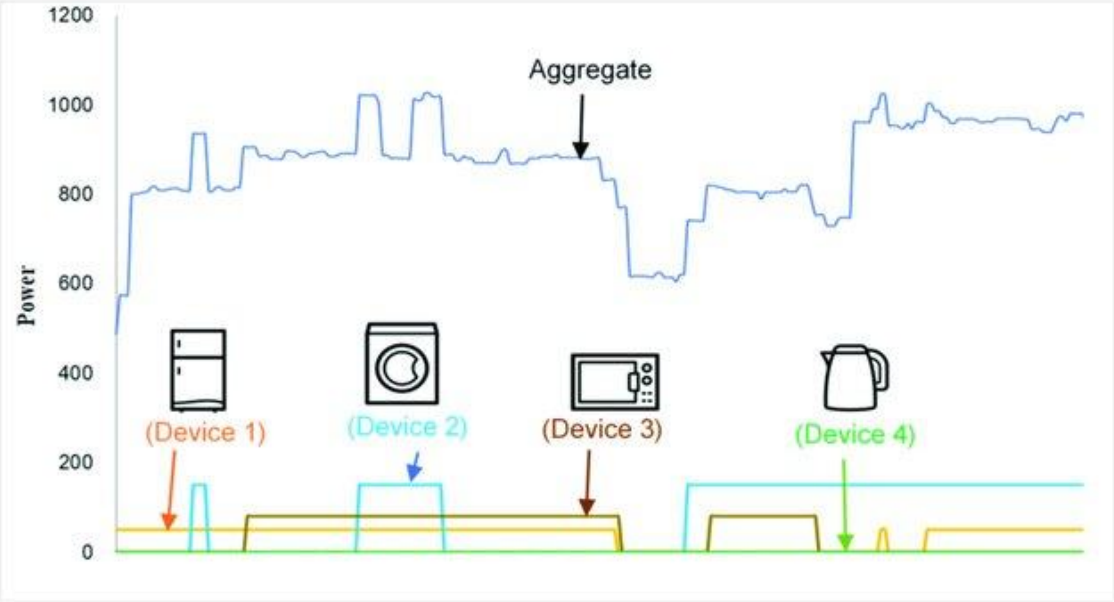
Labeling through simulation

- It is possible to rely on tools simulating consumption for each device
- Hence, individual simulated appliance consumption and their aggregation can be used to train models!
- **Drawback**:
    - challenging to have meta-information to simulate all the possible devices (e.g., different models, different operation modes)
    - synthetic data may not be realistic



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Mitigating labeled data scarcity with Transfer Learning

- It is possible to leverage ***transfer learning*** to adapt data from one or multiple domains where labeled data are available to an unlabeled target domain
- A specific domain adaptation loss is used to learn features that are *invariant* with respect to the domain!
  - goal: minimizing the differences in learned features across domains
- **Drawbacks:**
  - source and target domains should be similar or related
  - data distributions between source and target domains should be similar

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Privacy Aspects

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Energy consumption data

- The information about energy usage may reveal customer habits and behaviors
  - e.g., which appliances are used and when!
- Companies may perform mining on human behavioral patterns for their own advantages
  - e.g., targeted marketing, aggregate statistic for third parties
  - Lack of control on how these data are actually used
- Smart devices (e.g., smart meters) may also be attacked by malicious entities to manipulate energy costs and/or obtain sensitive information

Possible countermeasures

- **Encryption schemes:** achieve confidentiality by protecting data.
  - there are techniques allowing performing computation on encrypted data
- **Differential privacy:** introduce noise to make sure that energy data from a home is associated to a ***group*** of indistinguishable homes.
  - the information should be still useful to energy operator, avoiding to infer data from specific homes
- **Trusted platforms:** data analysis runs in trusted hardware that makes it impossible for outsiders to access data.

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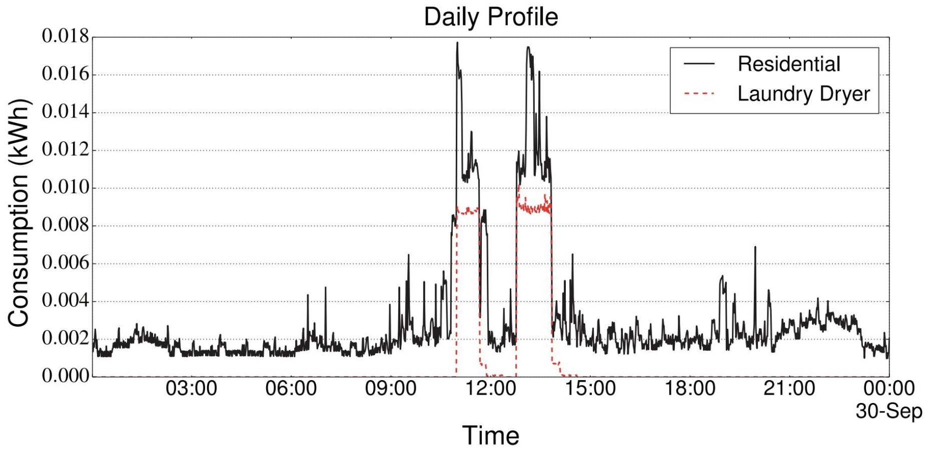
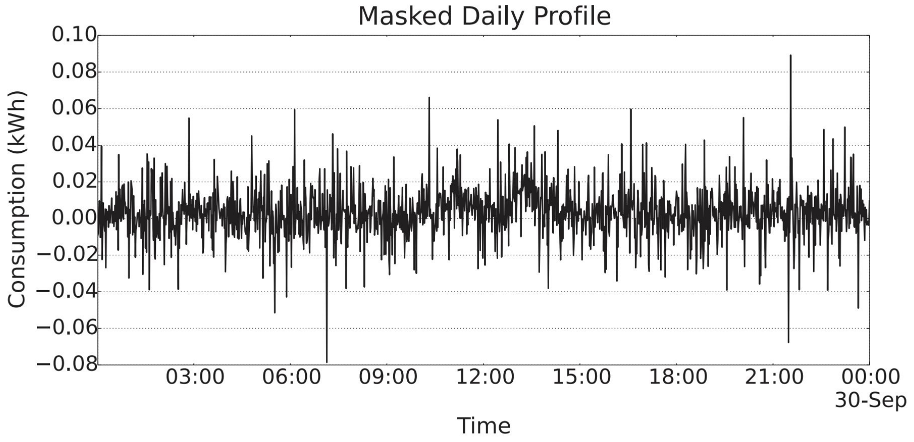
Example of a privacy-preserving strategy

- Goals:
  - Computing the total consumption of a consumer over a period of time (e.g., a month for billing)
  - Computing the total consumption of all consumers in a region at a certain instant of time
  - Avoiding to reveal the measurements of the instantaneous consumption of an individual consumer at a certain instant of time
- How? Perturbing consumption data with Laplacian noise (using a technique called “differential privacy”)





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